

MOHIT NIMJE

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SUMMARY

Computer Vision / Deep Learning practitioner with hands-on experience building CNN-based models for image analysis using PyTorch and OpenCV. Research background at BARC (RRCAT) applying deep learning to scientific imaging data (beam image parameter estimation), including image preprocessing, augmentation, and regression modeling.

EDUCATION

Bachelor of Technology in Computer and Science, VIT Bhopal, CGPA - 8.3 June 2023 - May 2027
Specialization - Artificial Intelligence and Machine Learning

SKILLS

Technical Skills	Python, PyTorch, OpenCV, TensorFlow, scikit-learn, NumPy, Pandas, Matplotlib
Tools And Concept	Computer Vision, Image Processing, CNNs, Image Segmentation, Feature Extraction, Deep Learning, Model Fine-Tuning
Soft Skills	Problem Solving, Quick Learner, Communication, Team Work

CERTIFICATIONS

• Applied Machine Learning in Python • Google IT Support

EXPERIENCE

Research Intern May 2025 - June 2025
Raja Ramanna Centre For Advanced Technology(BARC) *Indore, Madhya Pradesh*

- Designed and trained a Convolutional Neural Network (CNN) for beam image parameter estimation, achieving an approximate 0.14 validation MAE and strong regression performance.
- Built a custom image preprocessing and augmentation pipeline for scientific imaging data.
- Collaborated with cross-functional teams to integrate frameworks with accelerator beam diagnostics systems, reduced manual analysis efforts by approximately 30%.

PROJECTS

Web Trust Analyzer (Group Project) ([GitHub](#)) December 2025 - May 2026

- Built a five-factor weighted Trust Scoring Algorithm (Go + Redis + SQLite) that dynamically classifies IP behavior across 5 trust tiers, achieving 100% test accuracy across 7 attack scenarios (SQLi, DDoS, bot detection).
- Engineered a Redis caching layer cutting trust score latency from 4ms to under 0.5ms, and built TBAC middleware enforcing differentiated rate limits — doubling trusted-IP access while blocking malicious ones — across an 8-layer WAF pipeline.

Signature Forgery ([GitHub](#)) January 2025 - April 2025

- Engineered a Signature Verification application using an MLP neural network, achieving 92% accuracy in distinguishing genuine from forged signatures.
- Built an end-to-end ML pipeline with scikit-learn, OpenCV, and scikit-image for feature extraction and classification, improving prediction accuracy by 15%.

Prediction of Heart Health using Heart Beat ([GitHub](#)) September 2024 - December 2024

- Constructed a data-driven heartbeat sound classification platform using Random Forest, achieving 90% accuracy on .wav audio files through statistical feature extraction.
- Deployed a scalable Flask API that runs in real time and provides inference to users with an average response time of 2 seconds.